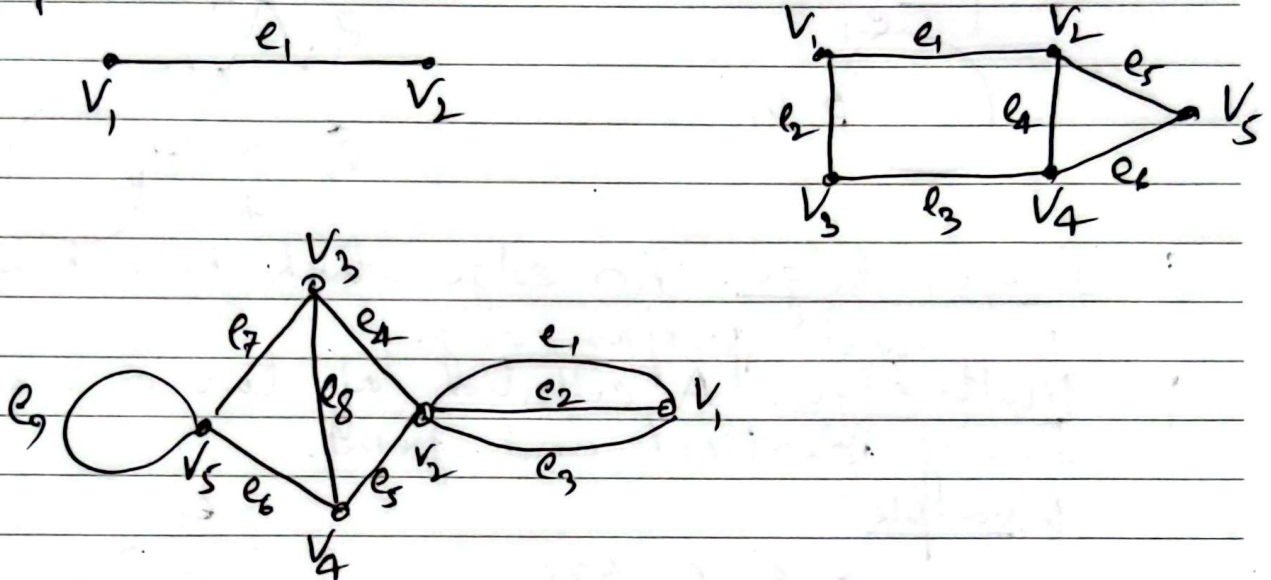


Graph

A graph G is a mathematical structure consisting of two sets V and E where V is a non-empty set of vertices and E is a non-empty set of edges.

Example:

Basic Terminology

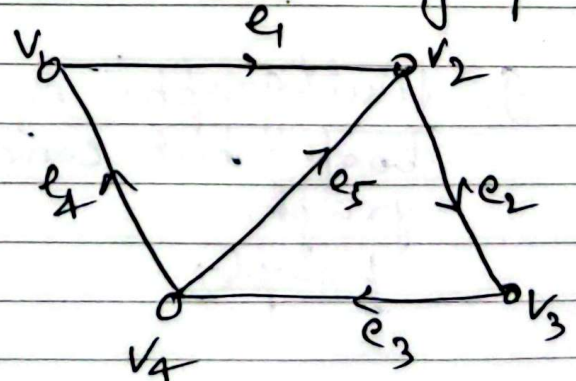
1. Trivial Graph: A graph consisting only one vertex and no edge.

e.g.

2. Null Graph: A graph consisting n vertices and no edge.

e.g.

3. Directed Graph: A graph consist the direction of edges then this is called directed graph.

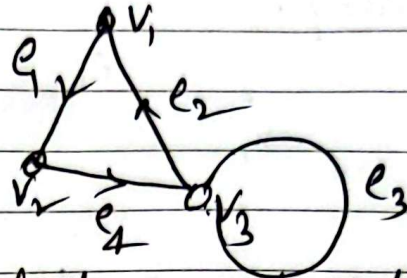
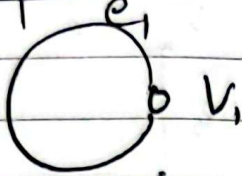


Notes

4. Undirected Graph: A graph which is not directed then it is called undirected graph.

5. Self loop in a Graph: If edge having the same vertex as both its ends vertices is called self loop.

Example:

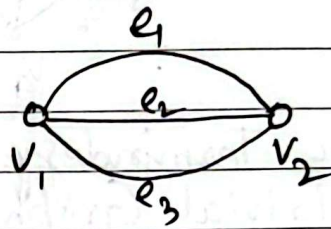


6. Proper Edge: An edge which is not self loop is called proper edge.

7. Multi edge: A collection of two or more edges having identical end-point.

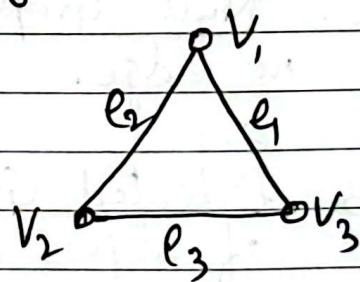
Example:

e_1, e_2, e_3 are multiedges.



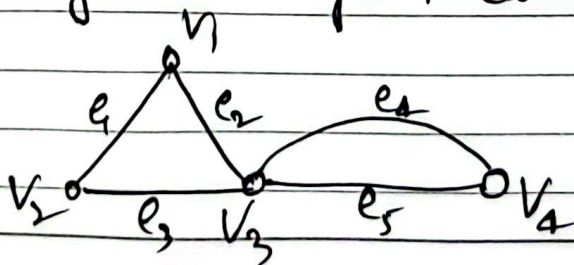
8. Simple Graph:

A graph not containing any self loop and multiedge.



9. Multigraph: A graph not containing any self loop but containing multiedge is called multigraph.

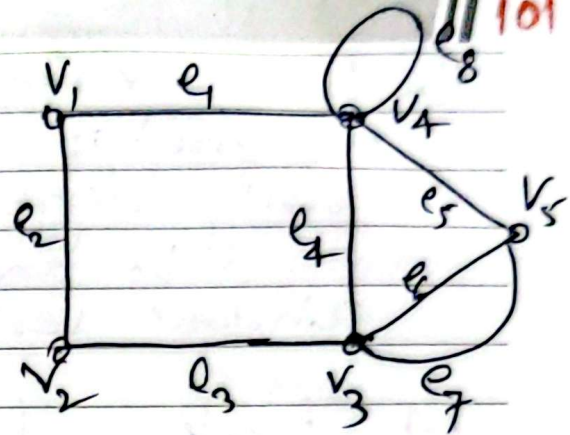
Example:



Notes

10. Pseudo Graph:

A graph containing both self loop and multiedge is called pseudo graph.

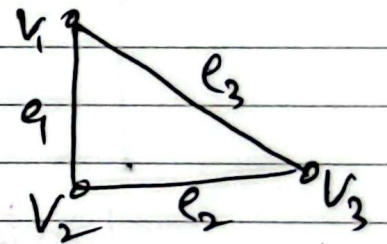


11. Incidence and Adjacency: Let e_k be an edge joining two vertices V_i and V_j , then e_k is said to be incident of V_i & V_j .

Two vertices are said to be adjacent if there exist an edge joining these vertices.

Example:

Here e_1 is incident of V_1 & V_2 . But V_1 and V_2 are adjacency for e_1 .



12. Degree of Vertex

The degree of vertex V in a graph G written as $d(V)$ is equal to the number of edges which are incident on V with self loop counted twice.

Example:

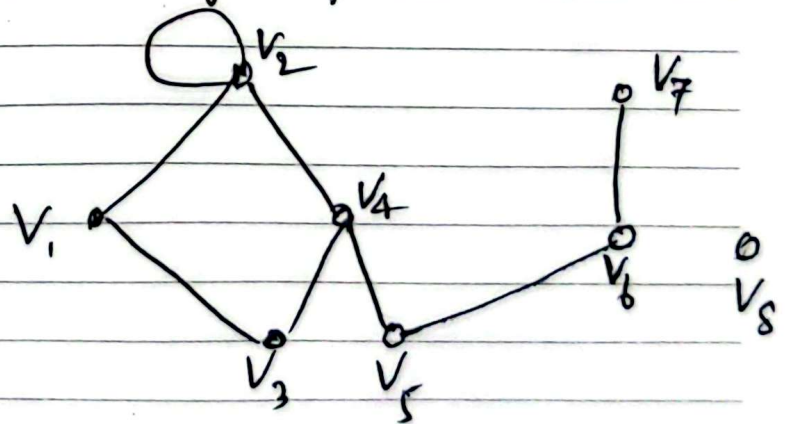
Here, $d(V_1) = 2$

$d(V_2) = 4$

$d(V_4) = 3$

$d(V_7) = 1$

$d(V_8) = 0$

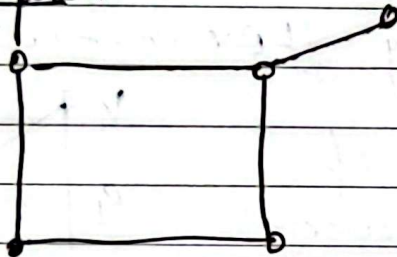


Notes

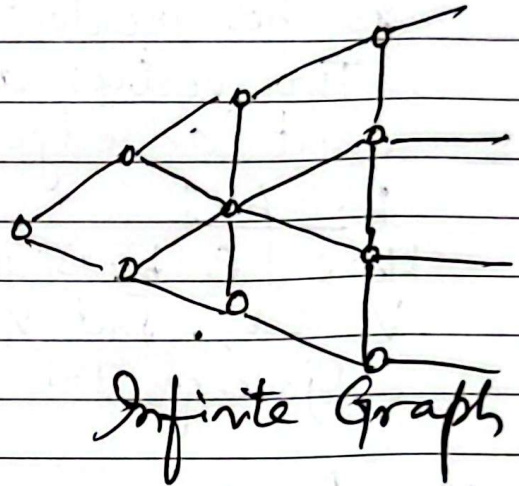
13. Isolated Vertex & Pendant Vertex: A vertex having 0 (degree) is called isolated vertex and a vertex having degree one is called pendant vertex.
e.g.: V_8 and V_7 are isolated vertex and pendant vertex respectively.

14. Finite and Infinite Graph: A graph with a finite number of vertices as well as edge is called finite graph otherwise it is an infinite graph.

Example:



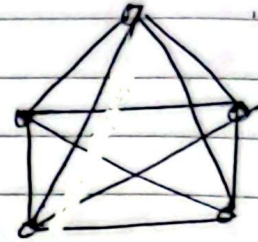
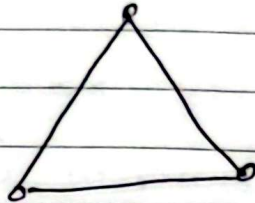
Finite Graph



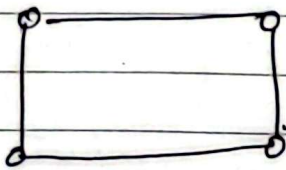
Infinite Graph

Some Important Graph

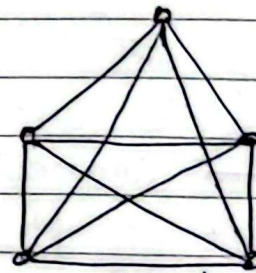
1. Complete Graph: A simple connected graph is said to be complete if each vertex is connected to every other vertex.



2. Regular Graph: A graph is said to be regular if every vertex has the same degree. If degree of each vertex of graph G is k , then it is called k -regular graph.

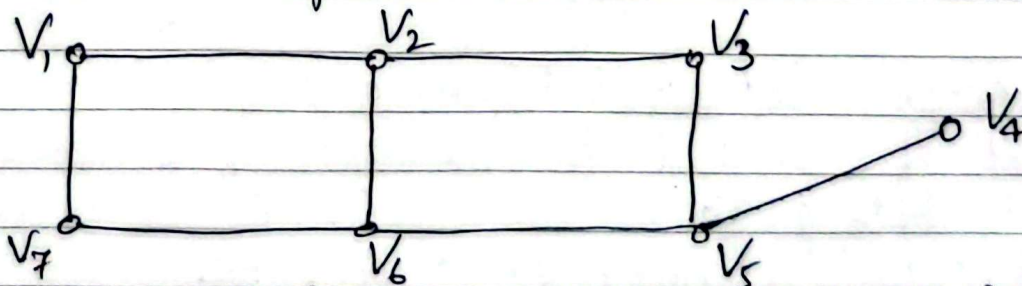


2- Regular Graph



4- Regular Graph

3. Bigraph (or Bipartite): If the vertex set V of a graph G can be partitioned into two non-empty disjoint subsets X and Y in such a way that edge of G has one end in X and one end in Y . Then G is called bipartite.

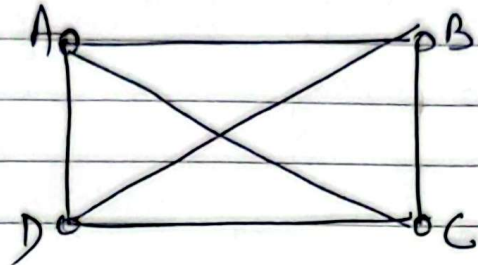


Here $V = \{V_1, V_2, V_3, V_4, V_5, V_6, V_7\}$

$X = \{V_1, V_2, V_3, V_4\}$ and $Y = \{V_7, V_6, V_5\}$.

Notes

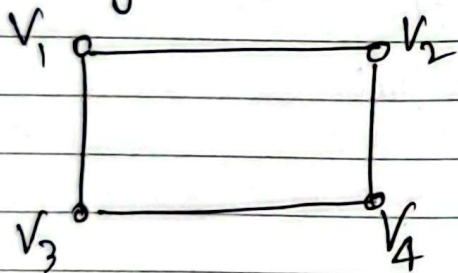
4. **Connected Graph**: An undirected graph is said to be connected if there is a path between every two vertices.



Note: If a graph is connected then it will not be bipartite.

(કિન્નમને યદિ $X = \{A\}$ લિફ્ઝો અને $Y = \{B, C, D\}$ હુન જાનુ ૨ યદિ $X = \{B\}$ લિફ્ઝો અને $Y = \{A, C, D\}$ હુન જાનુ ૧ યદિ same vertex i.e. B યાદિ (C, D) સેંગ પાનિ આડેડ અને કારિ B યાદિ (C, D) સેંગ આડેડેન)

5. **Complete Bipartite Graph**: If every vertex in X is disjoint connected with every vertex in Y in a bipartite graph, then it is called a complete bipartite graph. If X and Y contain m and n vertices respectively, then this graph is denoted by $K_{m,n}$.

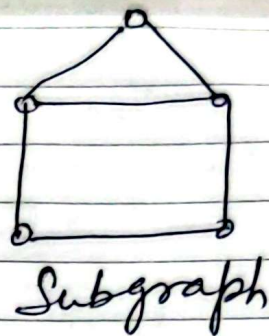
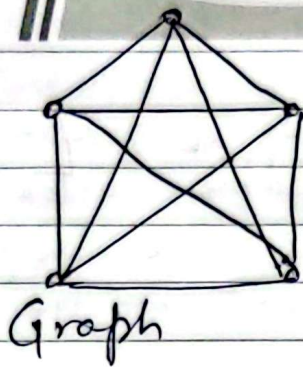


Here $X = \{V_1, V_4\}$ and $Y = \{V_3, V_2\}$.

Every vertex of X is connected with each and every vertex of Y .

It is denoted by $K_{2,2}$.

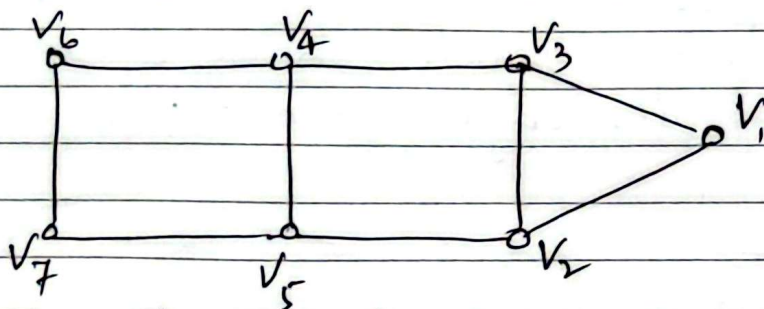
6. **Subgraph**: Let $G(V, E)$ be a graph. Let V' be a subset of V and E' be a subset of E whose end point belong to V' . Then $G(V', E')$ is a graph and called a subgraph of $G(V, E)$.



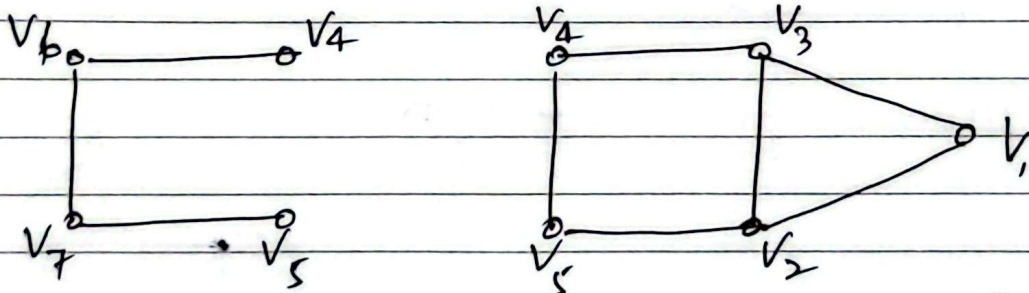
Decomposition of Graph

A graph is said to be decomposed into two subgraph G_1 and G_2 if

$$G_1 \cup G_2 = G \text{ and } G_1 \cap G_2 = \text{null graph.}$$



After decomposition:

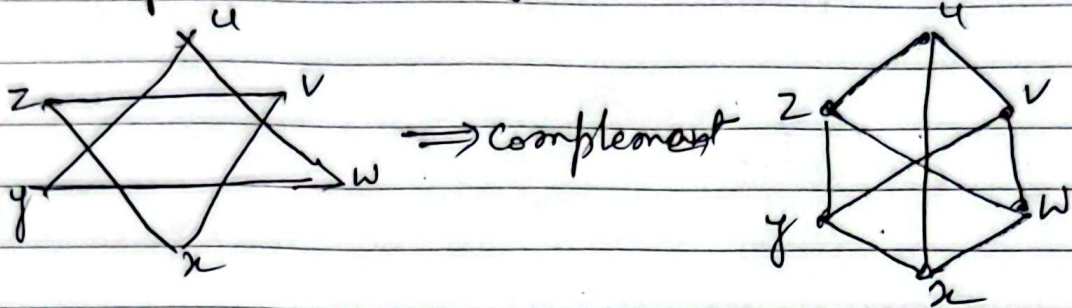


(यहाँ V_4 व V_5 दुई ग्राफ में दोबारा तापति नो दुई जोड़ने edge ~~है~~ पहिलो ग्राफ में है।)

Notes

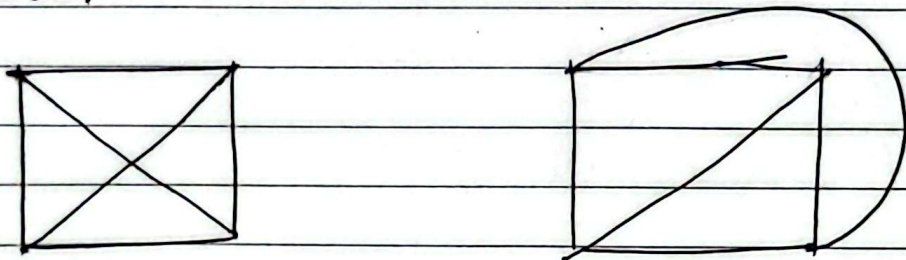
Complement of Graph

The complement of a graph G is defined a simple graph with the same vertex set as G and where two vertices u and v are adjacent only but they are not adjacent in G .

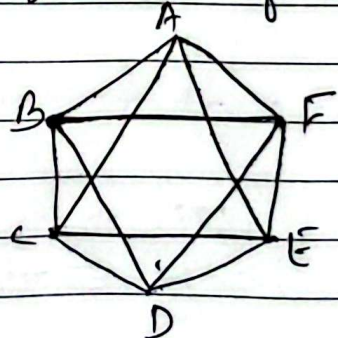


Planare Graph

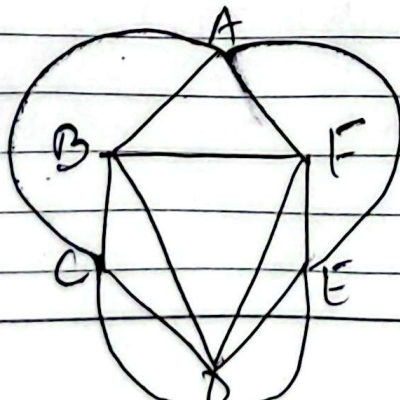
A graph which can be drawn in the plane so that its edges do not cross is called planare.



Que 1. Draw a planar graph representation of a graph.



Sol?:

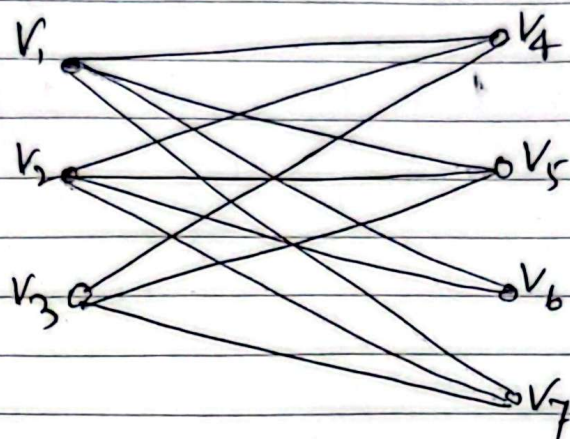


Notes

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Que 2: Draw $K_{3,4}$ bipartite graph.

Sol:



Handshaking Theorem

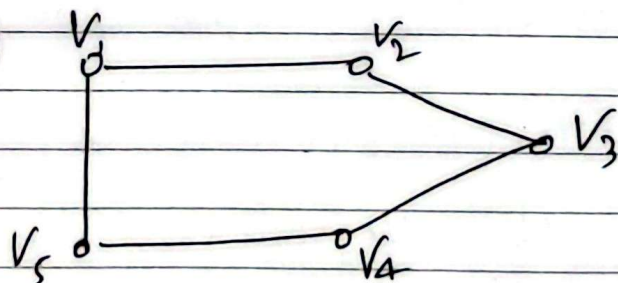
The sum of the degree of the vertices of a graph G is equal to twice the number of edges in G .

Matrix Representation of Graph

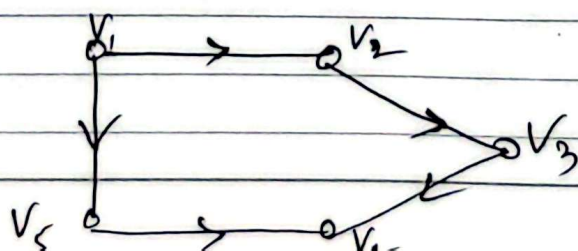
1. Adjacency Matrix: Let a_{ij} denote the number of edges (v_i, v_j) then $A = (a_{ij})_{n \times n}$ is called adjacency matrix of G if

$$a_{ij} = \begin{cases} 1 & \text{if } (v_i, v_j) \text{ is an edge.} \\ 0 & \text{otherwise.} \end{cases}$$

e.g. (a)



(b)



Notes

Their matrices are.

$$(a) \quad A = \begin{matrix} & \begin{matrix} v_1 & v_2 & v_3 & v_4 & v_5 \end{matrix} \\ \begin{matrix} v_1 \\ v_2 \\ v_3 \\ v_4 \\ v_5 \end{matrix} & \begin{bmatrix} 0 & 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 1 & 0 \end{bmatrix} \end{matrix}$$

$$(b) \quad A = \begin{matrix} & \begin{matrix} v_1 & v_2 & v_3 & v_4 & v_5 \end{matrix} \\ \begin{matrix} v_1 \\ v_2 \\ v_3 \\ v_4 \\ v_5 \end{matrix} & \begin{bmatrix} 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix} \end{matrix}$$

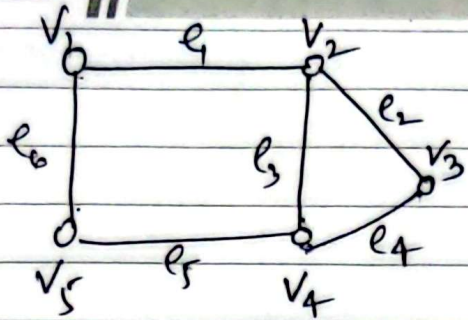
2. Incidence Matrix: Let G be a ~~matrix~~ graph with m vertices $v_1, v_2, \dots, v_m$ and n edges $e_1, e_2, \dots, e_n$.

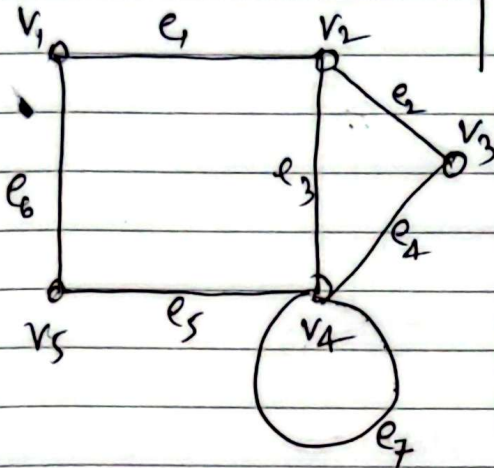
Let a matrix $M = (m)_{m \times n}$ defined by

$$m_{ij} = \begin{cases} 1 & \text{if } v_i \text{ is incident on } e_j \\ 0 & \text{if } v_i \text{ is not incident on } e_j \\ 2 & \text{if } v_i \text{ is an end of loop } e_j. \end{cases}$$

Notes

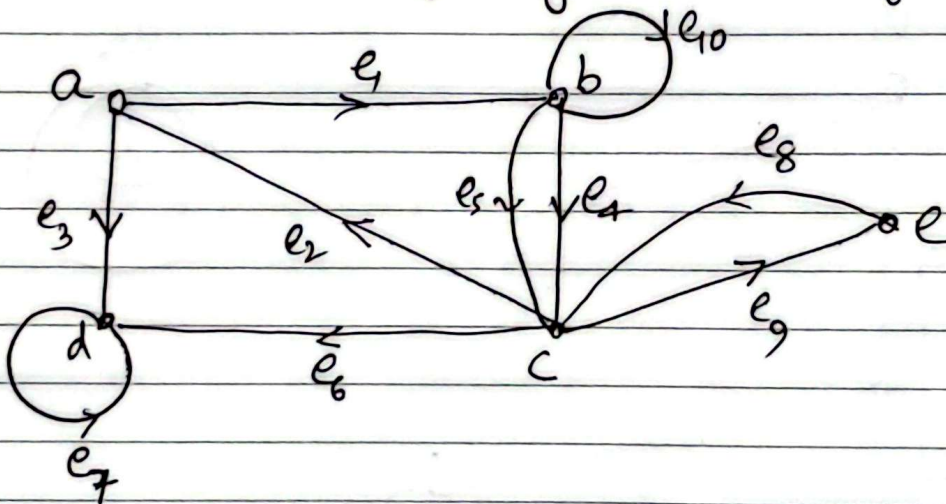
eg.



$$A = \begin{matrix} & \begin{matrix} e_1 & e_2 & e_3 & e_4 & e_5 & e_6 \end{matrix} \\ \begin{matrix} v_1 \\ v_2 \\ v_3 \\ v_4 \\ v_5 \end{matrix} & \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 \end{bmatrix} \end{matrix}$$


$$A = \begin{matrix} & \begin{matrix} e_1 & e_2 & e_3 & e_4 & e_5 & e_6 & e_7 \end{matrix} \\ \begin{matrix} v_1 \\ v_2 \\ v_3 \\ v_4 \\ v_5 \end{matrix} & \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 1 & 0 \\ 1 & 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 1 & 1 & 0 & 2 \\ 0 & 0 & 0 & 0 & 1 & 1 & 0 \end{bmatrix} \end{matrix}$$

Que 1. Find the Adjacency matrix of graph.



Sol?!

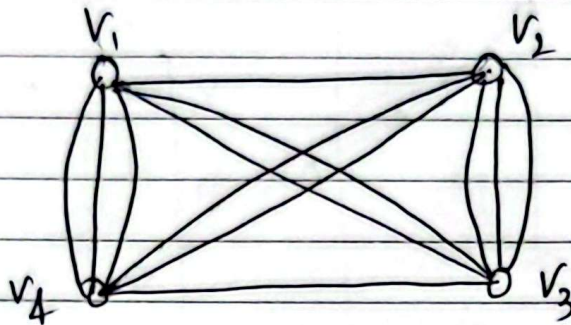
$$\begin{matrix} & \begin{matrix} a & b & c & d & e \end{matrix} \\ \begin{matrix} a \\ b \\ c \\ d \\ e \end{matrix} & \begin{bmatrix} 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 2 & 0 & 0 \\ 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 \end{bmatrix} \end{matrix}$$

Notes

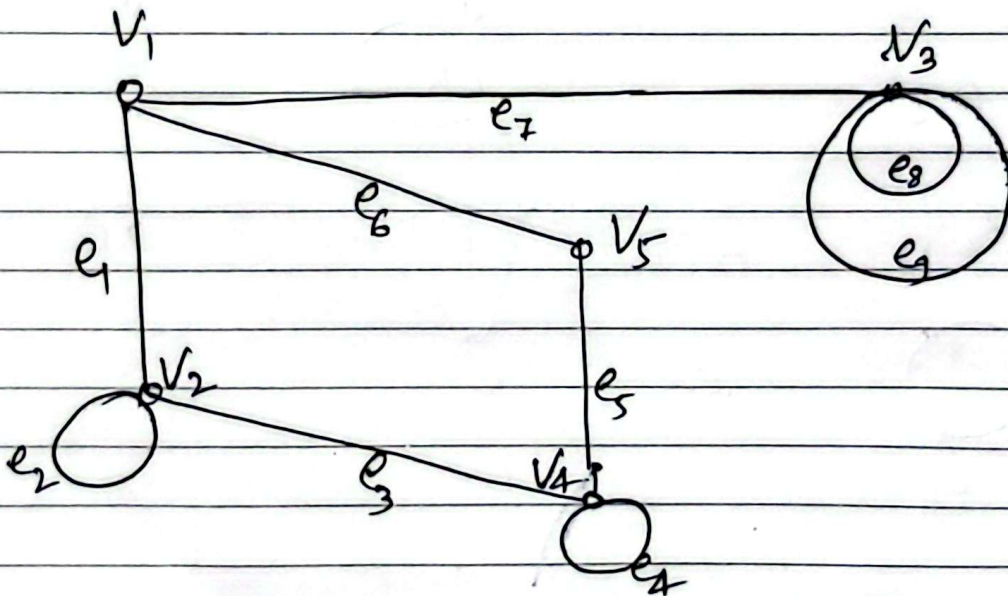
Que 2. A graph is given by the following adjacency matrix

$$A = \begin{matrix} & v_1 & v_2 & v_3 & v_4 \\ \begin{matrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{matrix} & \begin{pmatrix} 0 & 1 & 2 & 3 \\ 1 & 0 & 3 & 2 \\ 2 & 3 & 0 & 1 \\ 3 & 2 & 1 & 0 \end{pmatrix} \end{matrix}$$

Solⁿ:



Que 3. Write down the Adjacency and Incidence matrix of graph.



Solⁿ:

Adjacency matrix:

$$\begin{matrix} & v_1 & v_2 & v_3 & v_4 & v_5 \\ \begin{matrix} v_1 \\ v_2 \\ v_3 \\ v_4 \\ v_5 \end{matrix} & \begin{pmatrix} 0 & 1 & 1 & 0 & 1 \\ 1 & 1 & 0 & 1 & 0 \\ 1 & 0 & 2 & 0 & 0 \\ 0 & 1 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 & 0 \end{pmatrix} \end{matrix}$$

Notes

Incidence Matrix =

	e_1	e_2	e_3	e_4	e_5	e_6	e_7	e_8	e_9
V_1	1	0	0	0	0	1	1	0	0
V_2	1	2	1	0	0	0	0	0	0
V_3	0	0	0	0	0	0	1	2	2
V_4	0	0	1	2	1	0	0	0	0
V_5	0	0	0	0	1	1	0	0	0